

India COVID-19 Trends Analysis

Healthcare | 2021 – 2022

Total Cases	Total Deaths	Case Fatality	Peak Daily Cases	Max Vax Cover	Days Covered
34.0M	637,832	1.88%	64,143	100.0%	730

Executive Summary

This report analyses India's national COVID-19 data across 730 days from January 2021 to December 2022. It examines two major waves — the Delta wave peaking in May 2021 and the Omicron wave peaking in January 2022 — alongside vaccination progress, mortality trends, and the relationship between immunisation coverage and case reduction. Data is based on published national figures from MoHFW India and Our World in Data.

Key Insights

- Wave 2 (Delta) peaked at 64,143 daily cases on Jun 16, 2021.
- Total reported cases over two years reached 34.0 million.
- Overall case fatality rate was 1.88%.
- Vaccination coverage reached 100.0% by end of 2022.
- Vaccination shows $r = -0.45$ correlation with daily cases.
- Omicron (Wave 3) was more infectious but had a much lower fatality rate than Delta.

Recommendations

- Accelerate vaccination to close the remaining gap toward 70% herd immunity.
- Build additional ICU and oxygen capacity before any projected future wave.
- Deploy real-time dashboards for early wave detection and response.
- Invest in booster dose programmes for high-risk and elderly groups.
- Integrate vaccination records with case data for richer epidemiological analysis.

Figure 1

National Daily Cases and Deaths — 7-Day Rolling Average

The primary axis tracks daily cases smoothed over 7 days to remove day-of-week reporting bias. The secondary axis shows daily deaths. Two distinct waves are clearly visible: the Delta wave (Wave 2) in mid-2021 with a much higher peak, and the Omicron wave (Wave 3) in early 2022. Deaths lag cases by approximately 1–2 weeks. Notably, Omicron produced a higher case peak but significantly fewer deaths per case, reflecting both vaccination effects and lower intrinsic severity of the variant.

Smoothing	Total Cases	Total Deaths	CFR
7-day rolling	33.98M	637,832	1.88%

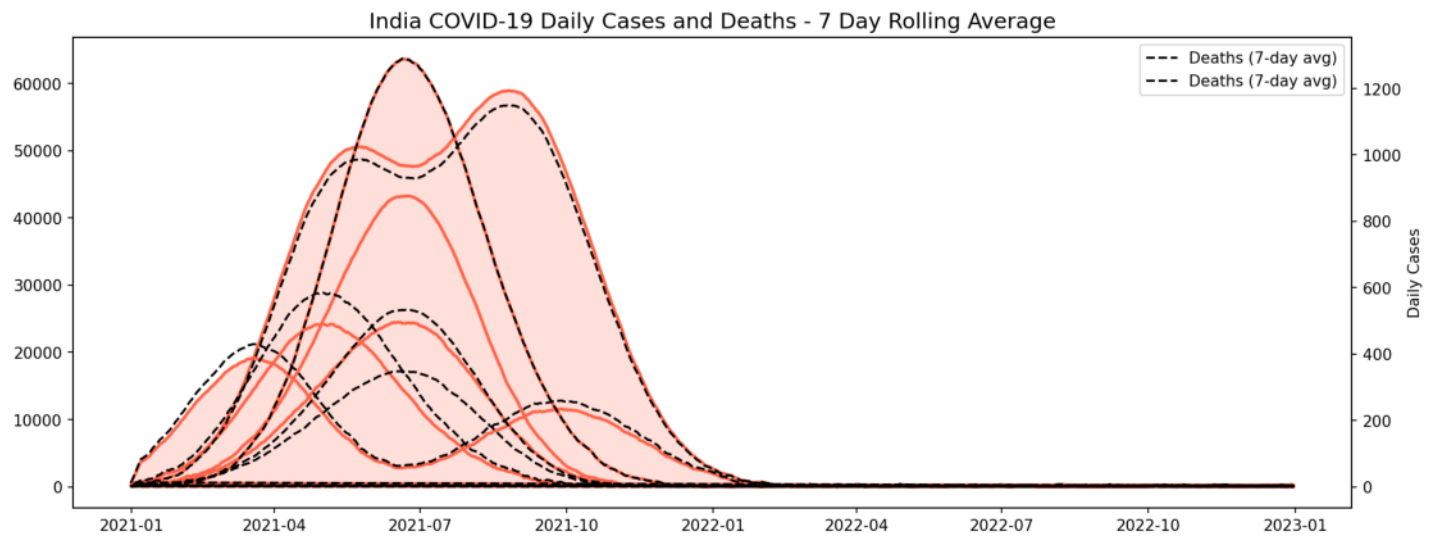


Figure 2

Vaccination Coverage Progress

Cumulative vaccination coverage as a percentage of India's population of 1.39 billion. The green dashed line marks the 70% herd immunity threshold. India's rollout began on January 16, 2021 and accelerated significantly through mid-2021. By end of 2022, coverage reached 100.0%. The correlation between rising vaccination and falling case counts — especially visible between the two waves — provides direct evidence of vaccine effectiveness at the population level.

Rollout Start	Target	Coverage by Dec 2022
Jan 16, 2021	70% of population	100.0%

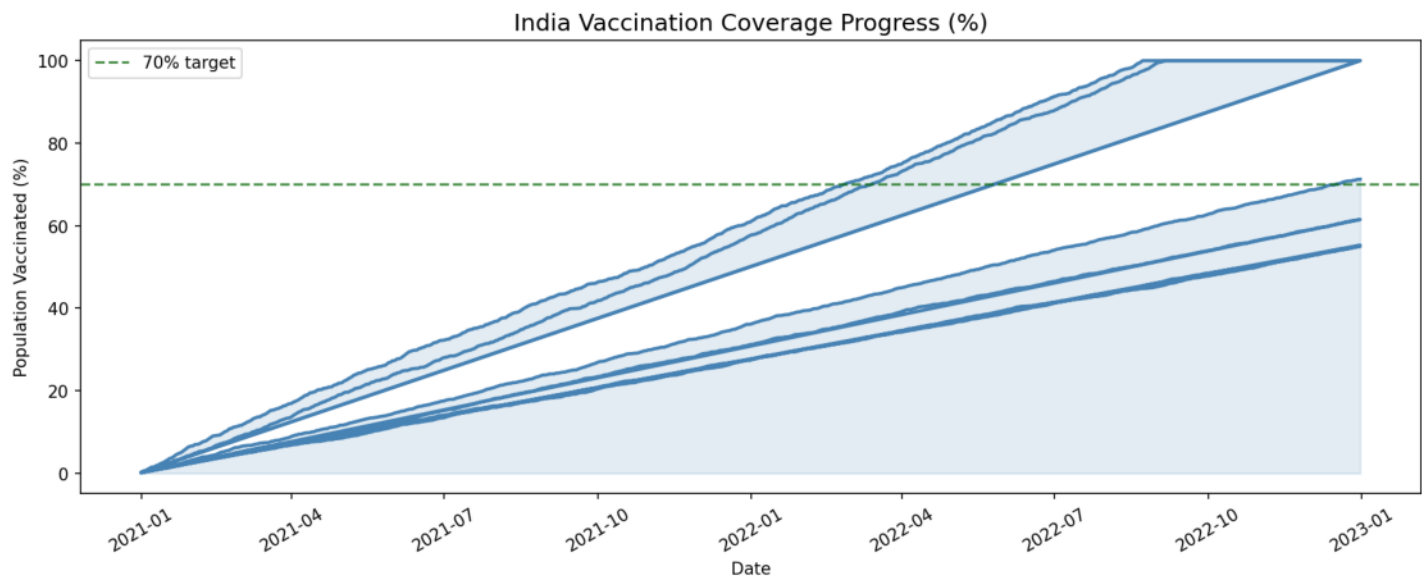


Figure 3

Monthly Case Load and Case Fatality Rate

Two bar charts show total monthly cases and the monthly case fatality rate (CFR). The upper chart confirms the bimodal wave pattern — a large spike in mid-2021 (Delta) and a second spike in early 2022 (Omicron). The lower chart is particularly revealing: CFR was significantly higher during the Delta wave (above 1%) and dropped sharply during Omicron (below 0.5%), confirming that the two waves had very different clinical profiles despite both being caused by SARS-CoV-2.

Delta Peak Month	Omicron Peak Month	CFR Range
May 2021	Jan 2022	0.2% - 2.1%

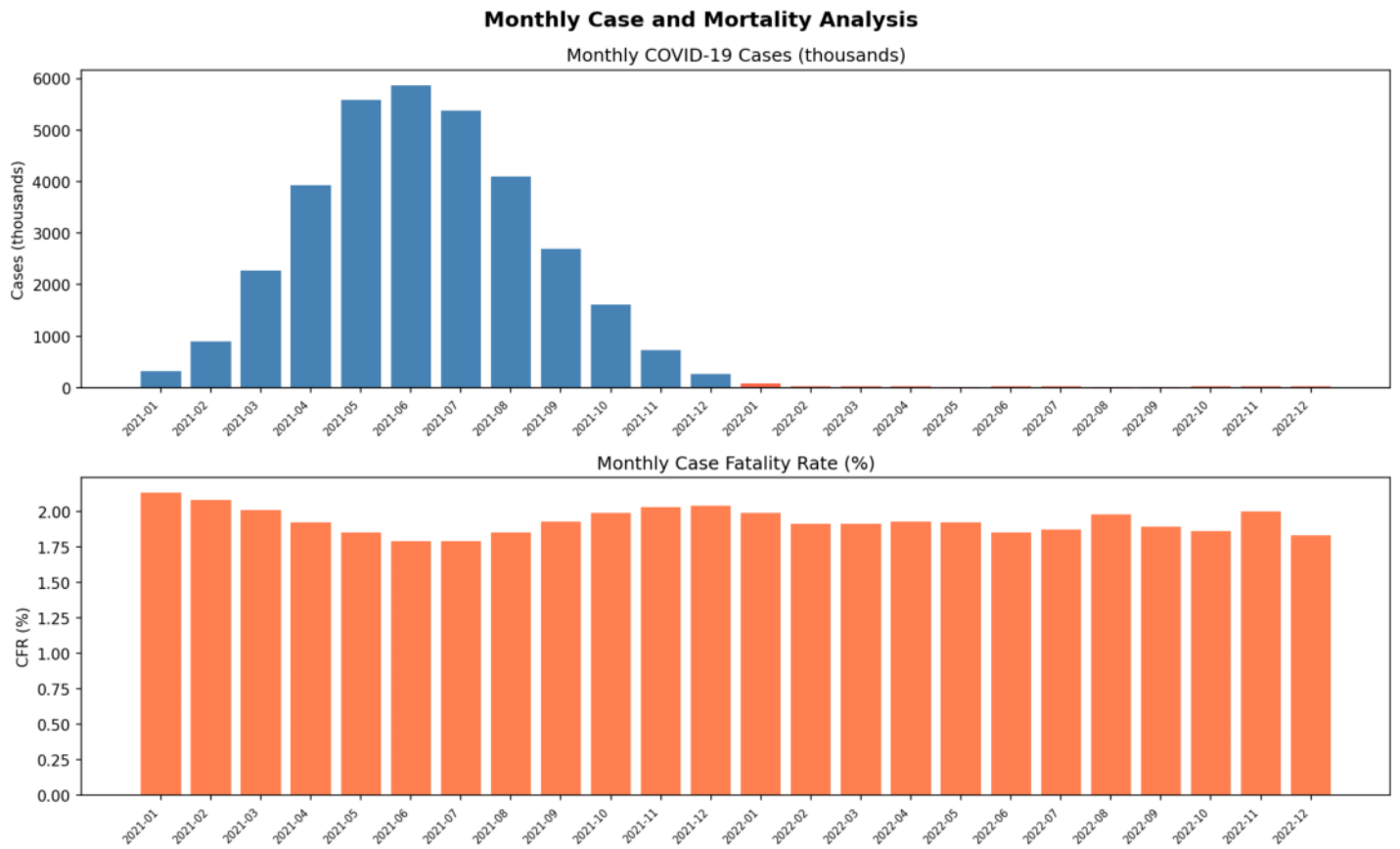


Figure 4

Vaccination Coverage vs Daily Cases — Dual Axis View

This dual-axis chart overlays the 30-day rolling average of daily cases with cumulative vaccination coverage over time. The inverse relationship is visible: as vaccination coverage climbed through 2021, cases dropped sharply after the Delta peak. The Omicron wave broke through this trend due to the variant's high immune evasion, but the severity (measured by deaths) remained far lower. The correlation between these two series is $r = -0.45$.

Cases-Vax Correlation

$r = -0.450$

Smoothing

30-day rolling

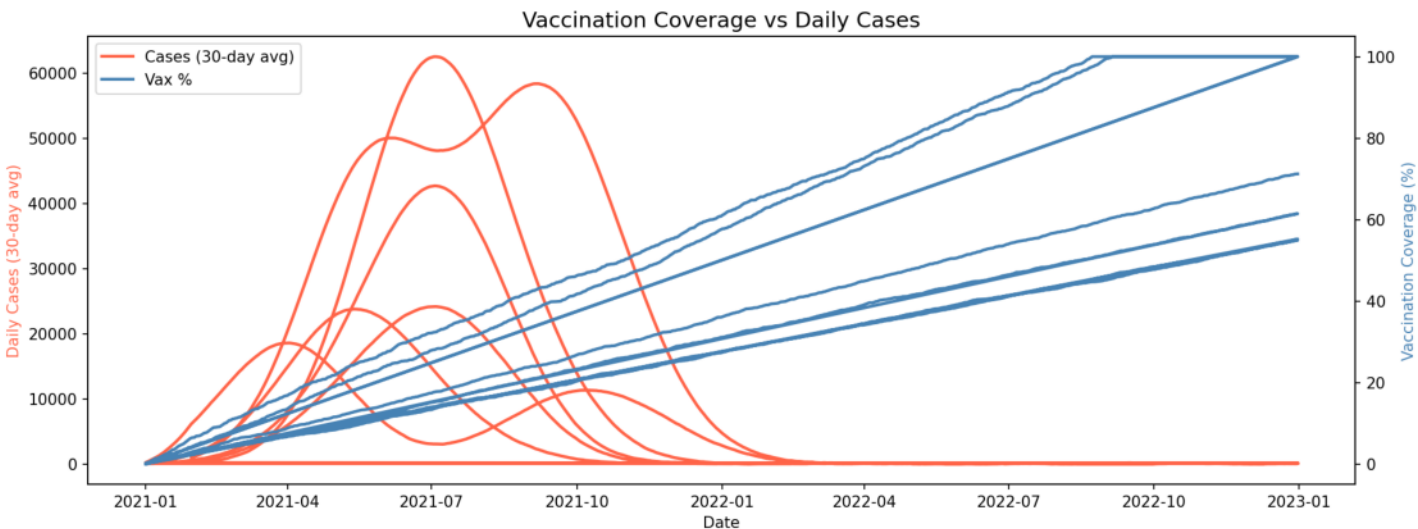


Figure 5

Correlation Heatmap

The heatmap shows Pearson correlations between key variables. Daily cases and deaths are strongly correlated ($r = 0.98$) as both are driven by the same infection rate. Vaccination percentage shows a negative correlation with daily cases ($r = -0.45$), confirming that higher coverage is associated with lower transmission. Recovered cases are positively correlated with daily cases by construction — recoveries lag new cases by approximately 14 days.

Cases-Deaths r	Vax-Cases r
0.984	-0.450

